

GPU* Architectures

Computing at Scale

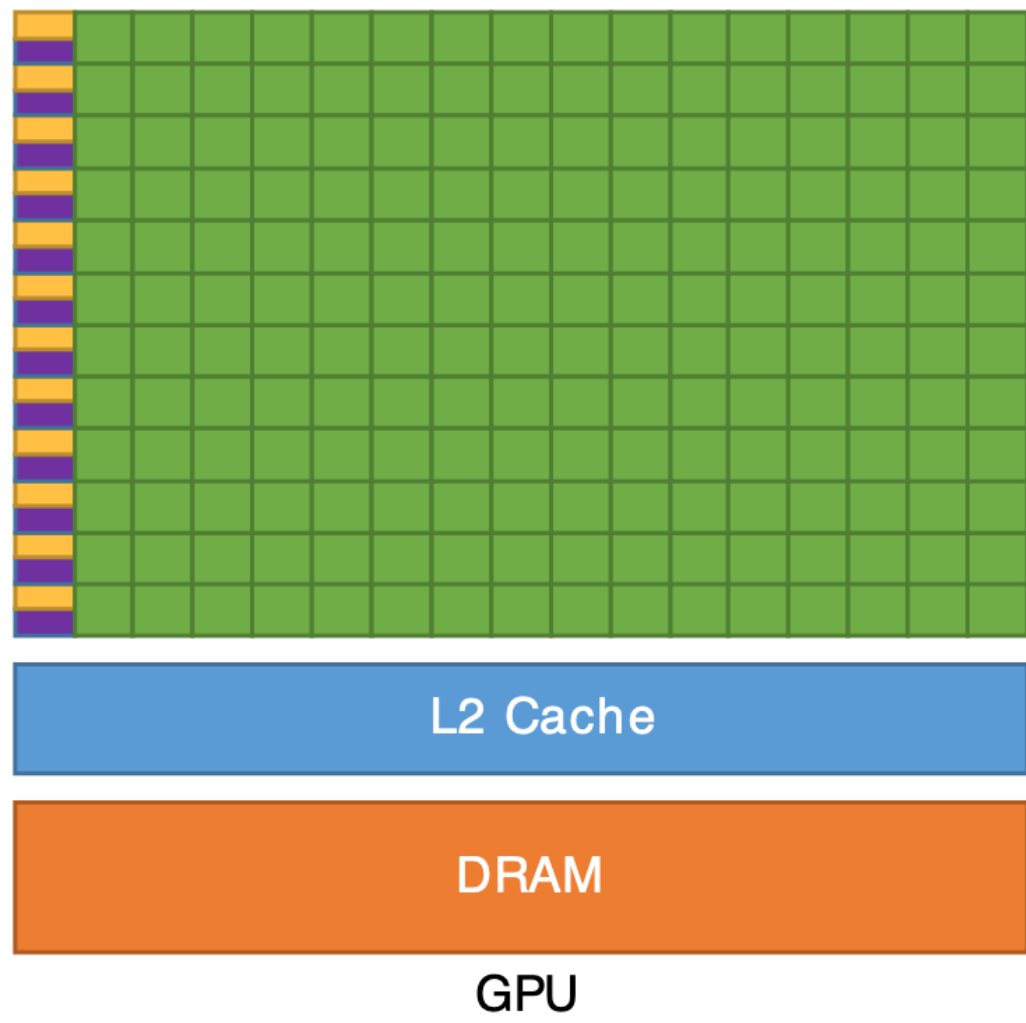
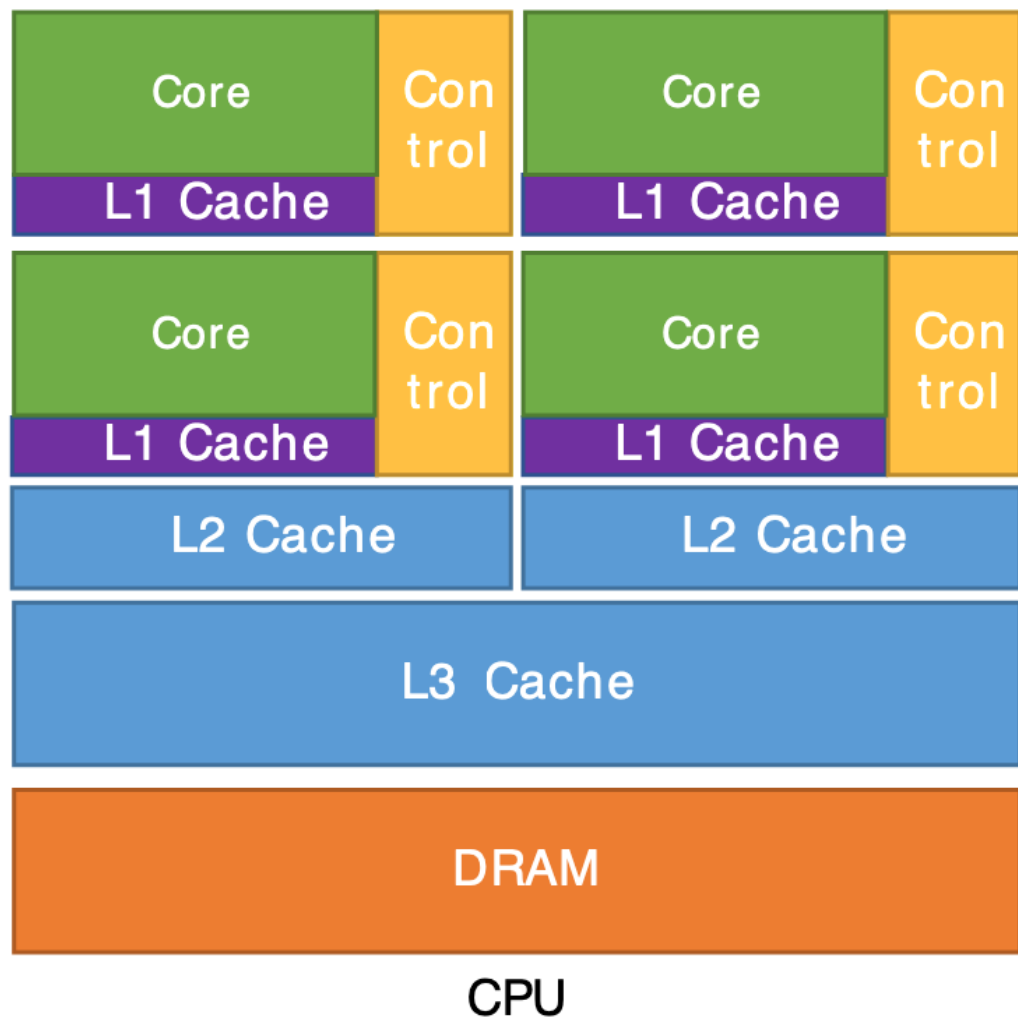
Max Hawkins

Key Topics

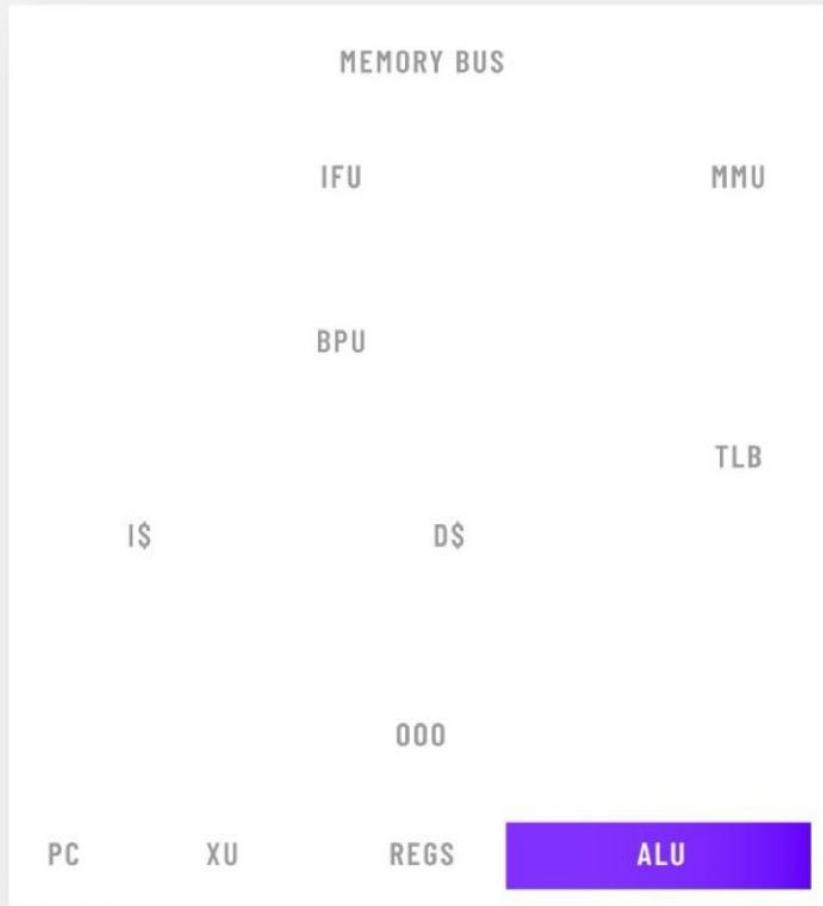
- Parallelism
- Dominance of Matrix Multiplication
- High-Bandwidth Memory
- Power
- GPU-GPU Interconnects
- Performance

Parallelism

- Sequential/Serial: Execute tasks one after the other
- Parallel: Execute many tasks simultaneously
- Many tasks are parallelizable
 - LLMs are very parallelizable
- GPUs are great at parallel execution
 - Many simple execution units
 - Maximizes throughput vs CPUs minimizing latency
- Latency hiding vs parallelism



Instruction Set Architecture



~2% of instruction set architectures are dedicated to compute (Arithmetic Logic Units - ALU) with the remainder being dedicated to overhead

NextSilicon Dataflow Architecture



The majority of NextSilicon's dataflow design is composed of ALUs, dramatically increasing compute potential

Apple Silicon

M4

28 billion transistors

N3E TSMC (2n gen. 3nm)

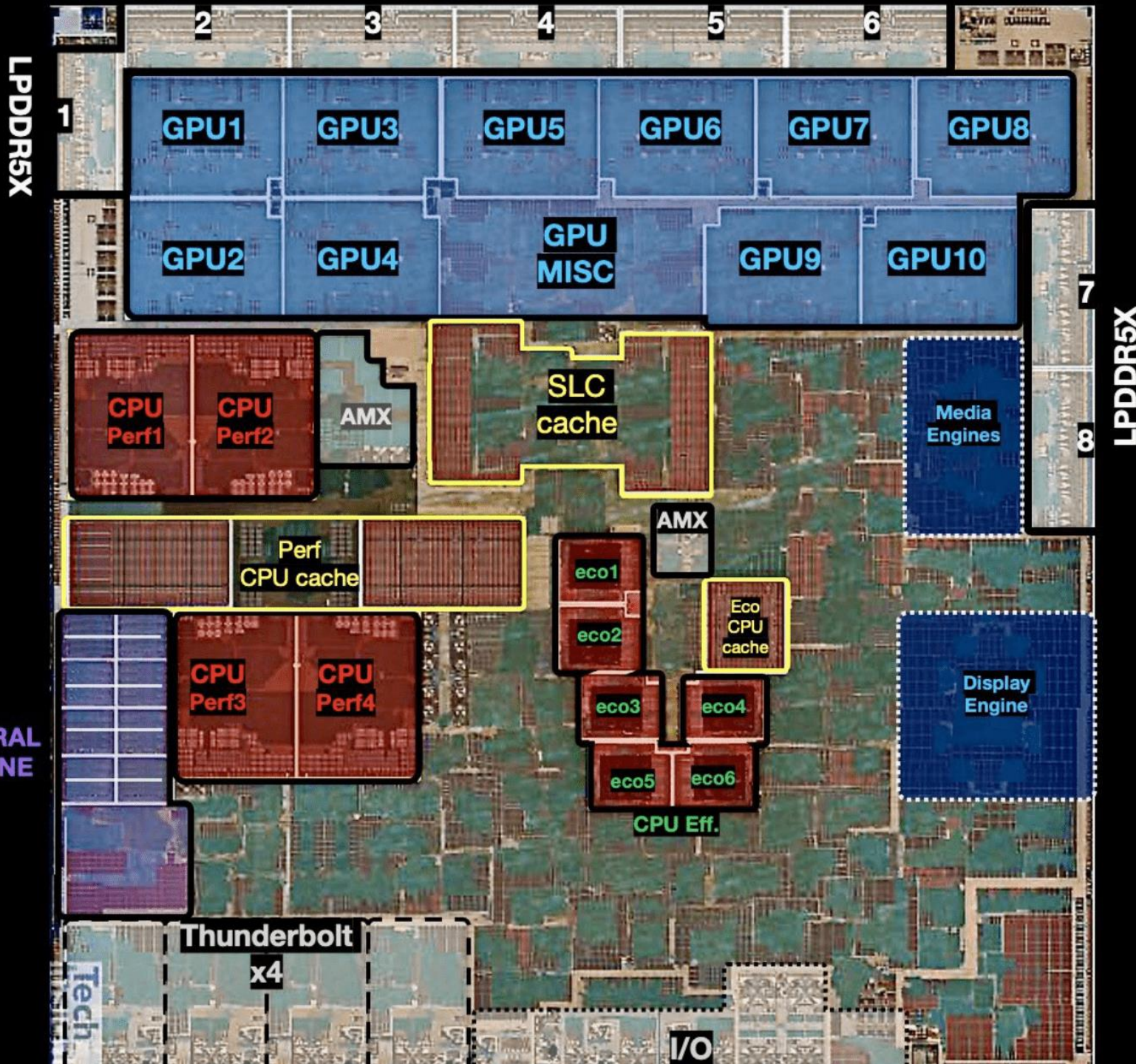
Die size : 1.28mm x 1,21mm

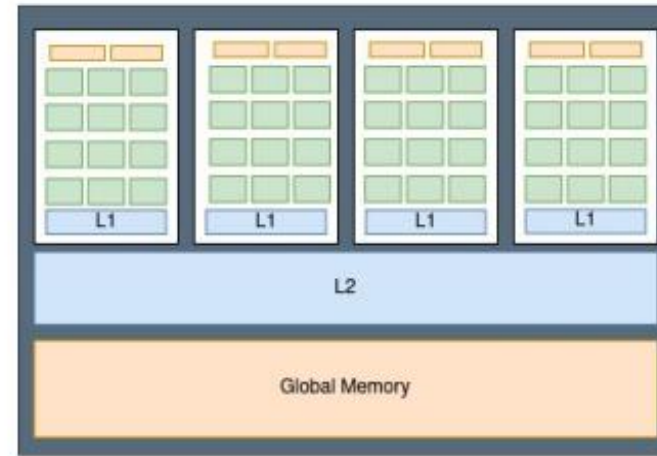
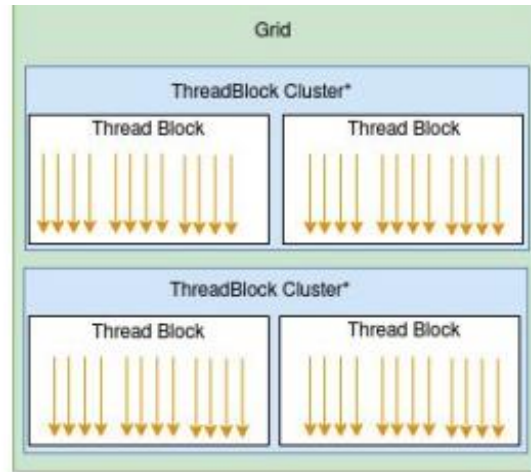


Frederic_Orange

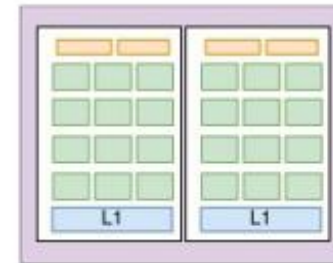
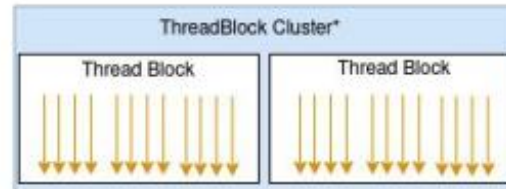


NEURAL
ENGINE

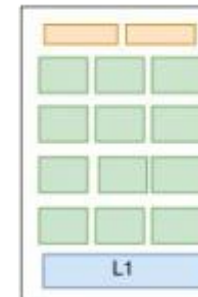
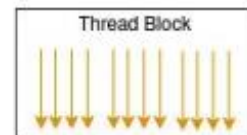




GPU Device
CPU is referred to as Host



Graphic Processing Cluster (GPC)



Streaming Multiprocessor



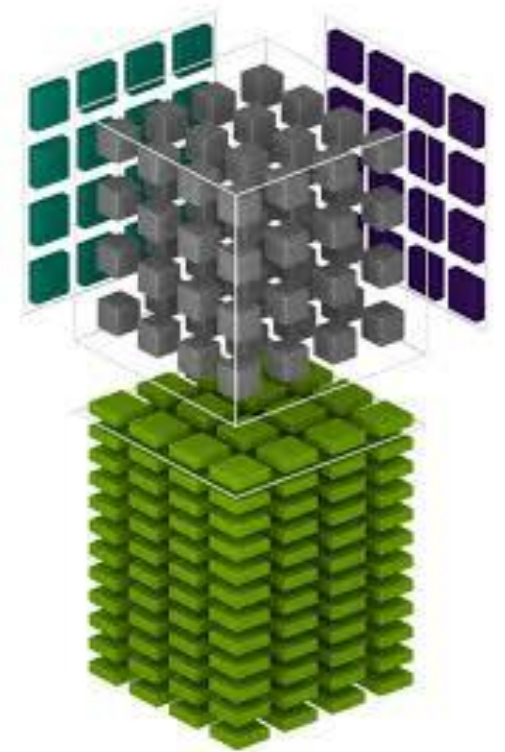
Thread

Matrix Multiplication Accelerators

Matrix operations underly *most* of modern LLM workflows

- Nvidia – Tensor Cores
- AMD – Matrix Cores
- Google – Tensor Processing Units (TPUs)
- Microsoft – Tile Tensor Unit (TTU on Maia 200)
- Apple – Neural Engine/AMX*

$2N^3$ operations: Power-intensive!



High-Bandwidth Memory (HBM)

- Von Neumann architectures:
 - Data movement and compute
- Tensor cores are computing units
- Need fast memory system too: HBM

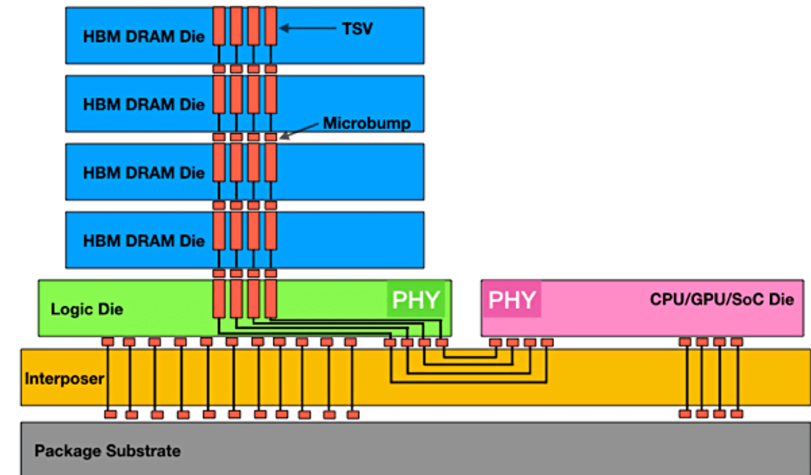
tom's **HARDWARE** premium

[PC Components](#) > [RAM](#)

Here's why HBM is coming for your PC's RAM — HBM consumes around three times the wafer capacity of DDR5 per gigabyte, as AI supercharges demand for chips and advanced packaging

Surging prices, diverted wafer capacity, and multi-year delays in relief are pushing memory out of reach for consumers.

Features By [Luke James](#) published December 19, 2025



Source: SemiconductorEngineering

Power

- Static power: Idle power caused by leakage currents*
 - Larger effects at smaller feature size
 - Power gating - 'unplug' a part of the chip if it's not being used
- Dynamic power: Consumed during transistor switching states
 - $O(C * V^2 * f)$
- More computation, bigger chips = More power

Technical Specifications	
	H200 SXM ¹
FP64	34 TFLOPS
FP64 Tensor Core	67 TFLOPS
FP32	67 TFLOPS
TF32 Tensor Core ²	989 TFLOPS
BFLOAT16 Tensor Core ²	1,979 TFLOPS
FP16 Tensor Core ²	1,979 TFLOPS
FP8 Tensor Core ²	3,958 TFLOPS
INT8 Tensor Core ²	3,958 TFLOPS
GPU Memory	141GB
GPU Memory Bandwidth	4.8TB/s
Decoders	7 NVDEC 7 JPEG
Confidential Computing	Supported
Max Thermal Design Power (TDP)	Up to 700W (configurable)
Multi-Instance GPUs	Up to 7 MIGs @18GB each
Form Factor	SXM
Interconnect	NVIDIA NVLink: 900GB/s PCIe Gen5: 128GB/s
Server Options	NVIDIA HGX™ H200 partner and NVIDIA-Certified Systems™ with 4 or 8 GPUs
NVIDIA AI Enterprise	Add-on

1. Preliminary specifications. May be subject to change.
2. With sparsity.

Individual Blackwell GPU Specific	
FP4 Tensor Core	20 petaFLOPS
FP8/FP6 Tensor Core	10 petaFLOPS
INT8 Tensor Core	10 petaOPS
FP16/BF16 Tensor Core	5 petaFLOPS
TF32 Tensor Core	2.5 petaFLOPS
FP32	80 teraFLOPS
FP64/FP64 Tensor Core	40 teraFLOPS
GPU Memory Bandwidth	186GB HBM3e 8 TB/s
Multi-Instance GPU (MIG)	
Decompression Engine	
Decoders	
Max Thermal Design Power (TDP)	Configurable up to 1,200W
Interconnect	5th Gen PCI
Server Options	NVIDIA GB200 NVL72 partner a NVIDIA-Certified Systems™ with 72

1. Preliminary specifications, subject to change. All Tensor Core numbers except FP64 with sparsi
2. Supported formats provide these speed-ups over H100 Tensor Core GPUs: 2X H.264, 1.25X HEV AV1 support is new to Blackwell GPUs.

2.3 kW	NVIDIA Rubin GPU
Configuration	1 NVIDIA Rubin GPU
NVFP4 Inference	50 PFLOPS
NVFP4 Training ²	35 PFLOPS
FP8/FP6 Training ²	17.5 PFLOPS
INT8 ²	0.25 POPS
FP16/BF16 ²	4 PFLOPS
TF32 ²	2 PFLOPS
FP32	130 TFLOPS
FP64	33 TFLOPS
FP32 SGEMM ³	400 TFLOPS
FP64 DGEMM ³	200 TFLOPS
GPU Memory Bandwidth	288 GB HBM4 22 TB/s
NVLink Bandwidth	3.6 TB/s

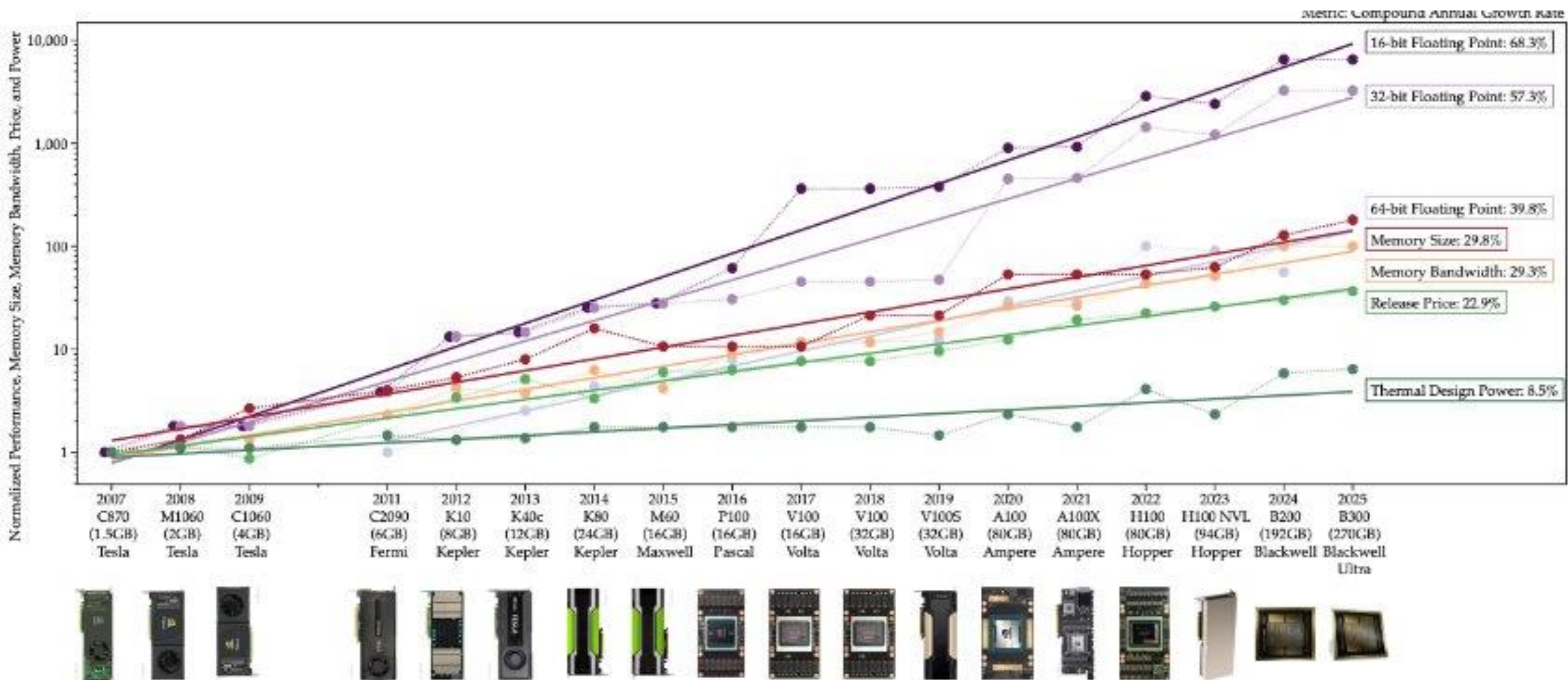


Image sources: ahead-it.eu, epoch.ai, nvidia.com, techpowerup.com, videocardz.net
 Price sources: lambdalabs.com, microway.com, northflank.com, shi.com, techpowerup.com, videocardz.net

Source: <https://arxiv.org/pdf/2601.20115>

GPU-GPU Interconnects

- Dedicated communication links amongst GPUs
- Nvlink, Infinity Fabric, UALink

